

Calculus I Lesson 28

1) $f(x) = x^3 - 6x^2 + 11x - 6$

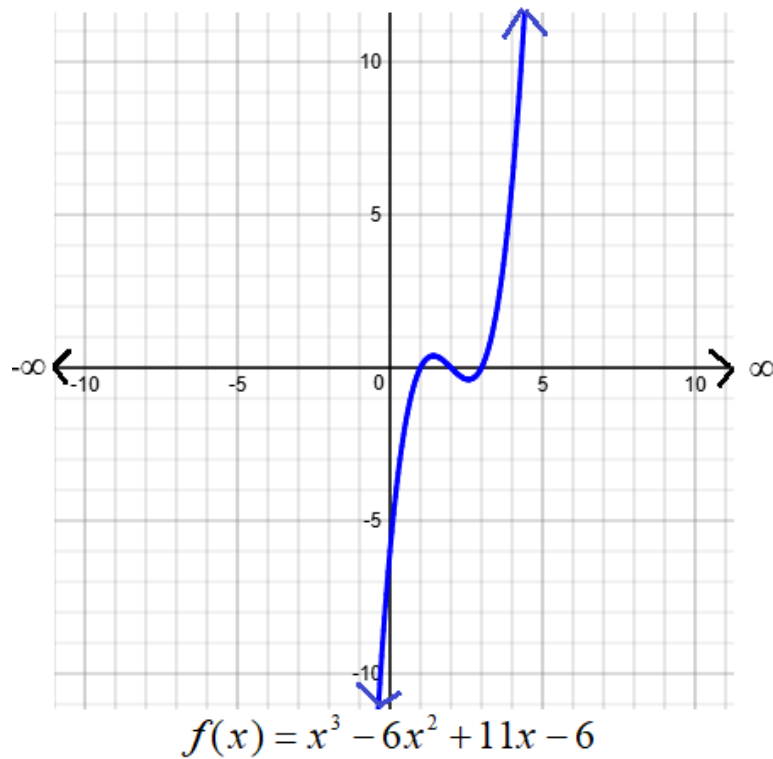
a) $f'(x) = \underline{\hspace{1cm}}?$

b) $f''(x) = \underline{\hspace{1cm}}?$

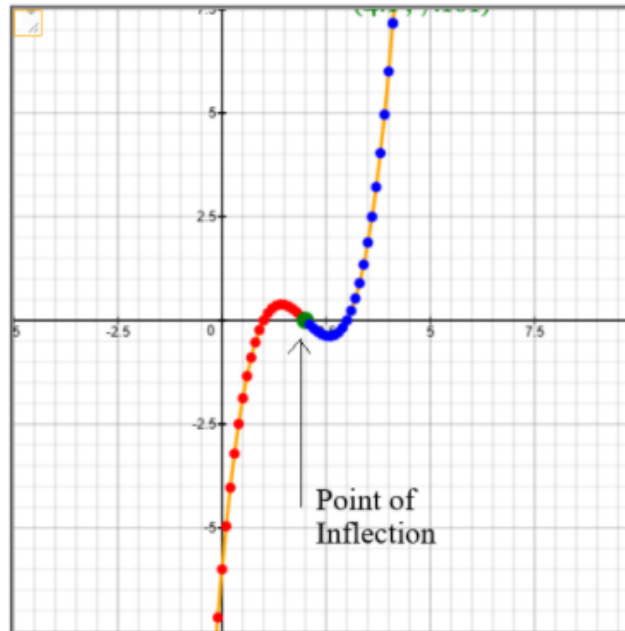
c) Set $f''(x) = 0$ and solve for x . $x = \underline{\hspace{1cm}}?$

d) The function $f(x) = x^3 - 6x^2 + 11x - 6$ concaves up on the interval: $\underline{\hspace{1cm}}?$

e) The function $f(x) = x^3 - 6x^2 + 11x - 6$ concaves down on the interval: $\underline{\hspace{1cm}}?$



Concavity and Point of Inflection



● $f''(x) < 0$; 2nd derivative is negative

● $f''(x) > 0$; 2nd derivative is positive

● Point of Inflection

2) $f(x) = x^3 - 2x^2 - 5x + 6$

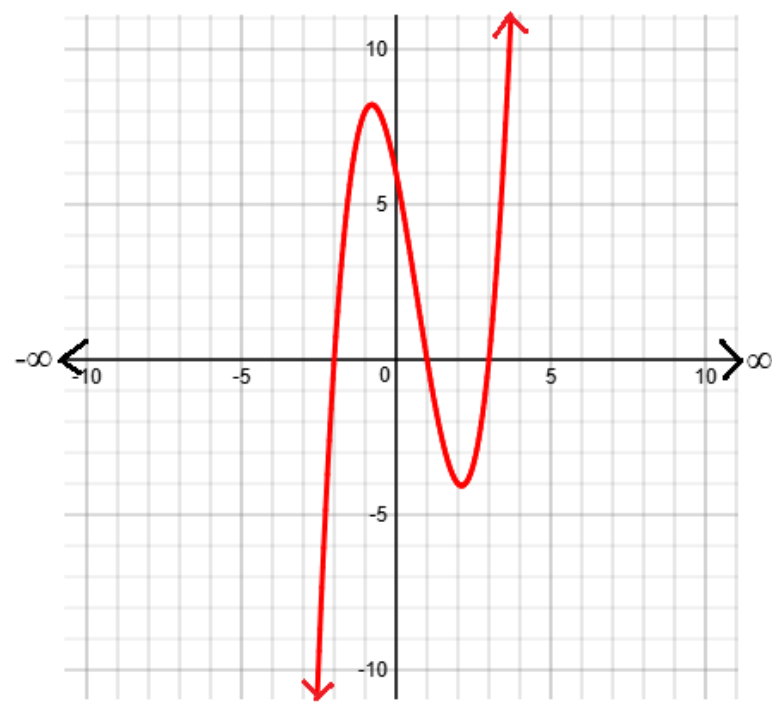
a) $f'(x) = \underline{\hspace{1cm}}?$

b) $f''(x) = \underline{\hspace{1cm}}?$

c) Set $f''(x) = 0$ and solve for x . $x = \underline{\hspace{1cm}}?$

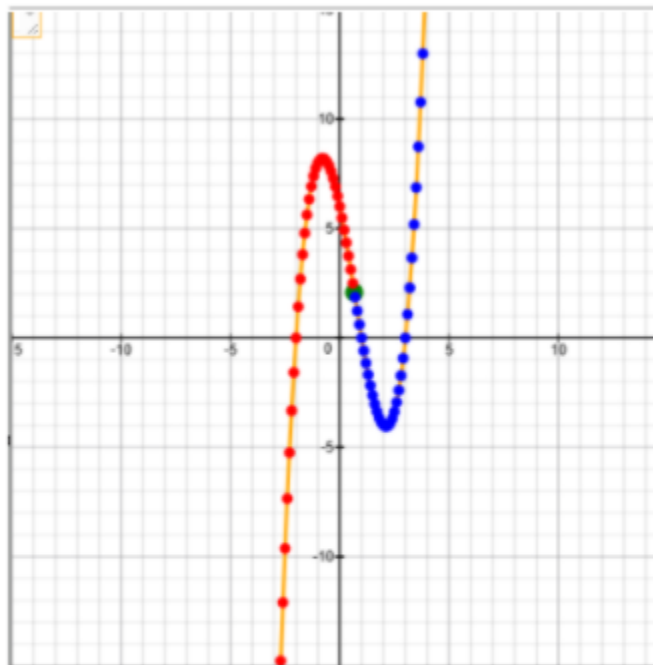
d) The function $f(x) = x^3 - 2x^2 - 5x + 6$ concaves up on the interval: $\underline{\hspace{1cm}}?$

e) The function $f(x) = x^3 - 2x^2 - 5x + 6$ concaves down on the interval: $\underline{\hspace{1cm}}?$



$f(x) = x^3 - 2x^2 - 5x + 6$

Concavity and Point of Inflection



● $f''(x) < 0$; 2nd derivative is negative

● $f''(x) > 0$; 2nd derivative is positive

● Point of Inflection

3) $f(x) = -0.5x^3 - 0.5x^2 + 7x + 12$

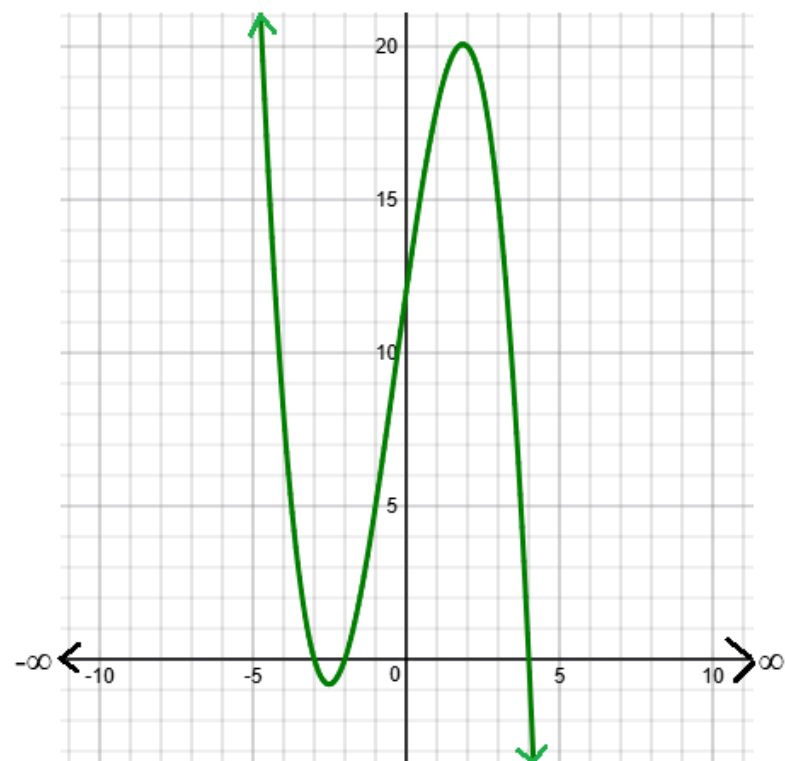
a) $f'(x) = \underline{\hspace{1cm}} ?$

b) $f''(x) = \underline{\hspace{1cm}} ?$

c) Set $f''(x) = 0$ and solve for x . $x = \underline{\hspace{1cm}} ?$

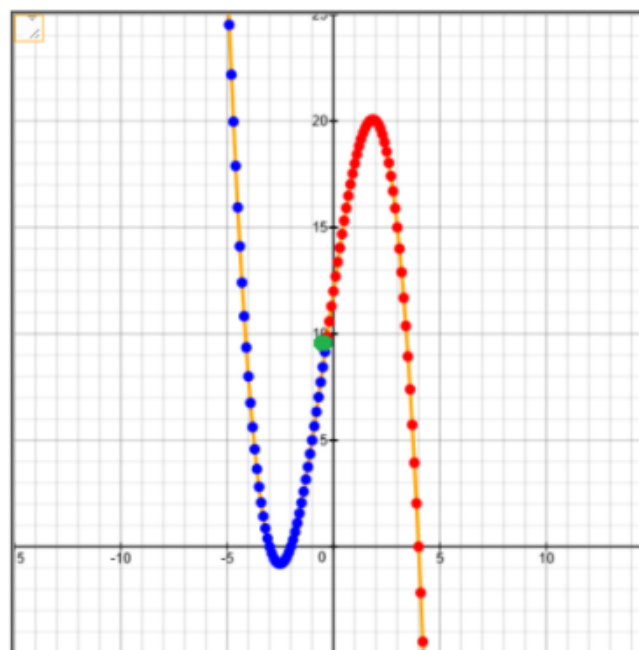
d) The function $f(x) = -0.5x^3 - 0.5x^2 + 7x + 12$ concaves up on the interval: $\underline{\hspace{1cm}} ?$

e) The function $f(x) = -0.5x^3 - 0.5x^2 + 7x + 12$ concaves down on the interval: $\underline{\hspace{1cm}} ?$



$$f(x) = -0.5x^3 - 0.5x^2 + 7x + 12$$

Concavity and Point of Inflection



● $f''(x) < 0$; 2nd derivative is negative

● $f''(x) > 0$; 2nd derivative is positive

● Point of Inflection

4) $f(x) = -0.5x^4 + 7.5x^2 + 5x - 12$

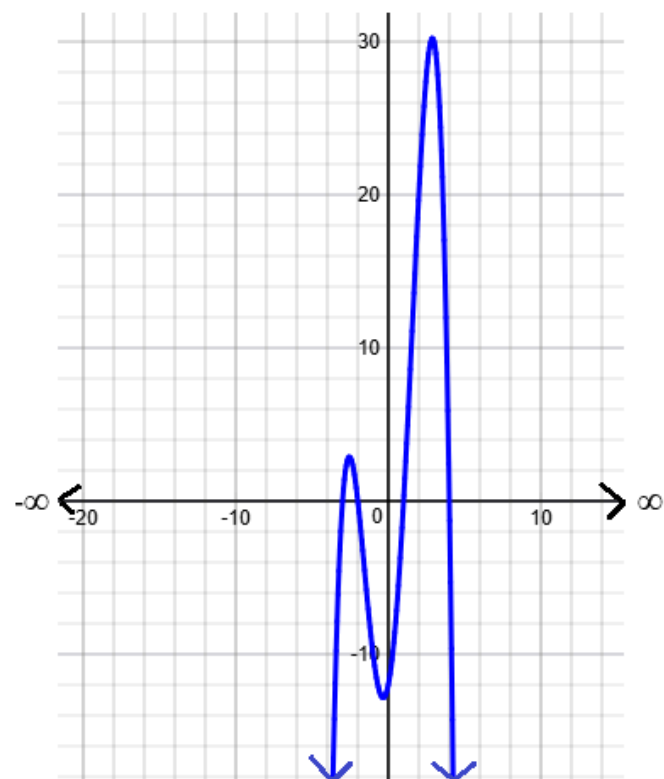
a) $f'(x) = \underline{\hspace{2cm}}?$

b) $f''(x) = \underline{\hspace{2cm}}?$

c) Set $f''(x) = 0$ and solve for x . $x = \underline{\hspace{2cm}}?$

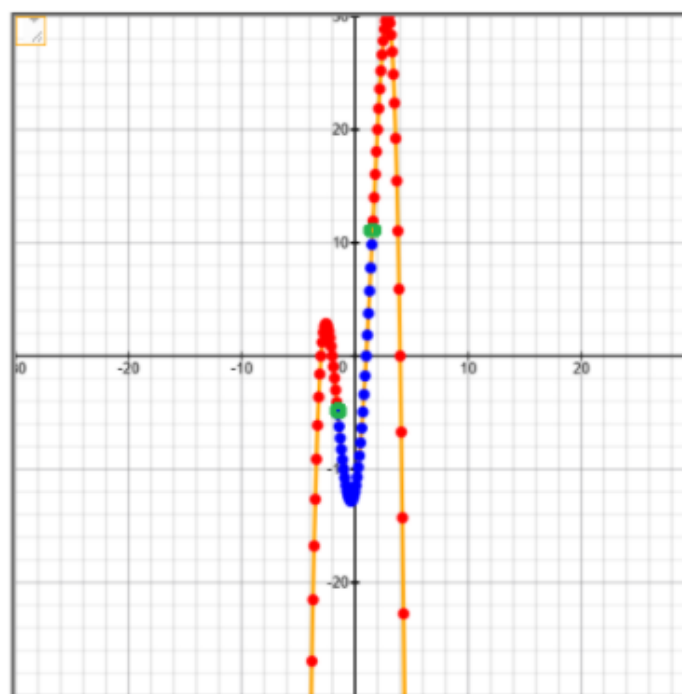
d) The function $f(x) = -0.5x^4 + 7.5x^2 + 5x - 12$ concaves up on the interval: $\underline{\hspace{2cm}}?$

e) The function $f(x) = -0.5x^4 + 7.5x^2 + 5x - 12$ concaves down on the interval: $\underline{\hspace{2cm}}?$



$f(x) = -0.5x^4 + 7.5x^2 + 5x - 12$

Concavity and Point of Inflection



● $f''(x) < 0$; 2nd derivative is negative

● $f''(x) > 0$; 2nd derivative is positive

● Point of Inflection

5) $f(x) = x^{2/3} + 1$

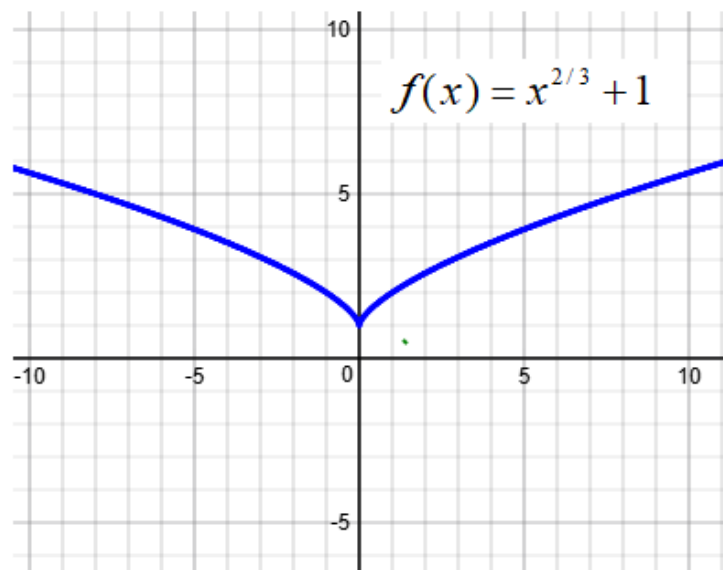
a) $f'(x) = \underline{\hspace{1cm}} ? \underline{\hspace{1cm}}$

b) $f''(x) = \underline{\hspace{1cm}} ? \underline{\hspace{1cm}}$

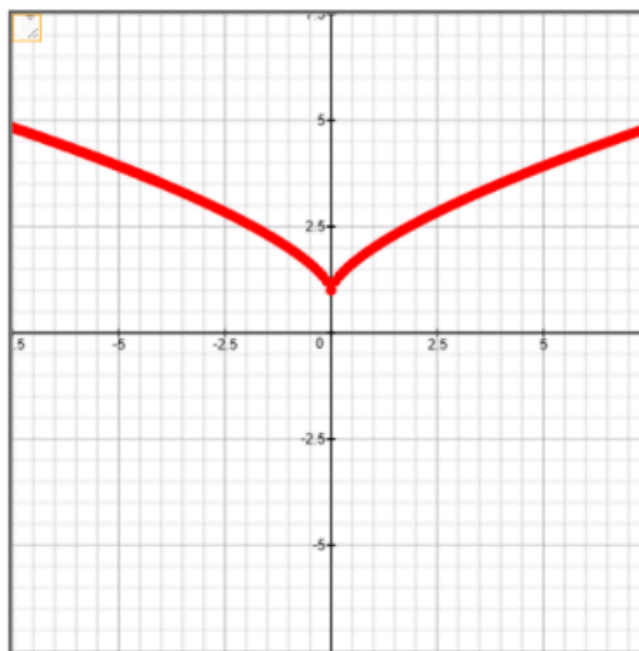
c) Set $f''(x) = 0$ and solve for x . $x = \underline{\hspace{1cm}} ? \underline{\hspace{1cm}}$.

d) The function $f(x) = x^{2/3} + 1$ concaves up on the interval: $\underline{\hspace{1cm}} ? \underline{\hspace{1cm}}$

e) The function $f(x) = x^{2/3} + 1$ concaves down on the interval: $\underline{\hspace{1cm}} ? \underline{\hspace{1cm}}$



Concavity and Point of Inflection



● $f''(x) < 0$; 2nd derivative is negative

● $f''(x) > 0$; 2nd derivative is positive

● Point of Inflection

6) $f(x) = x - 2\cos x$, $0 < x < 2\pi$

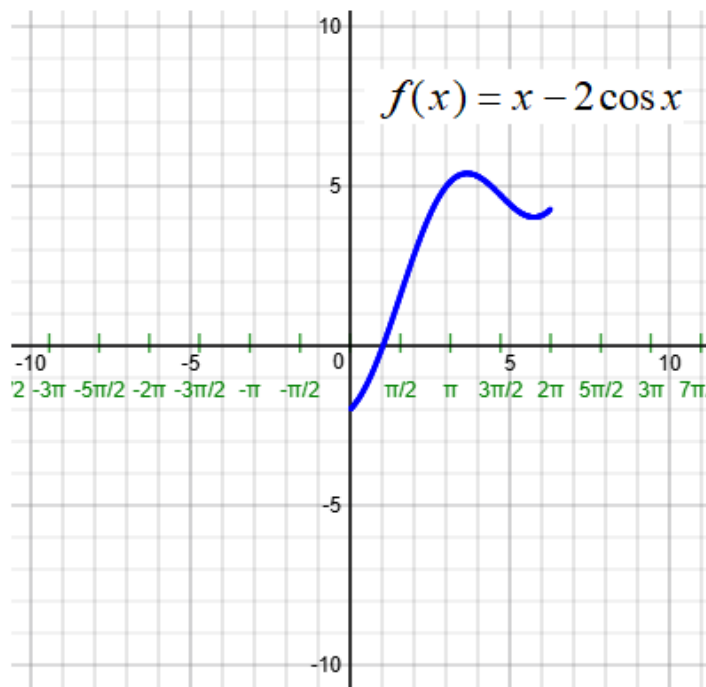
a) $f'(x) = \underline{\hspace{1cm}} ? \underline{\hspace{1cm}}$

b) $f''(x) = \underline{\hspace{1cm}} ? \underline{\hspace{1cm}}$

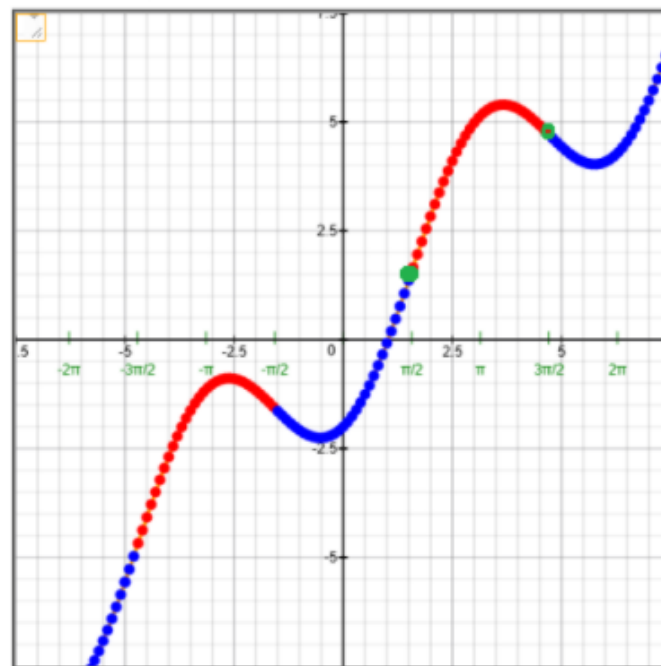
c) Set $f''(x) = 0$ and solve for x . $x = \underline{\hspace{1cm}} ? \underline{\hspace{1cm}}$.

d) The function $f(x) = x - 2\cos x$ concaves up on the interval: $\underline{\hspace{1cm}} ? \underline{\hspace{1cm}}$

e) The function $f(x) = x - 2\cos x$ concaves down on the interval: $\underline{\hspace{1cm}} ? \underline{\hspace{1cm}}$



Concavity and Point of Inflection



● $f''(x) < 0$; 2nd derivative is negative

● $f''(x) > 0$; 2nd derivative is positive

● Point of Inflection

7) $f(x) = \frac{1}{2}x + \sin x$, $0 \leq x \leq 2\pi$

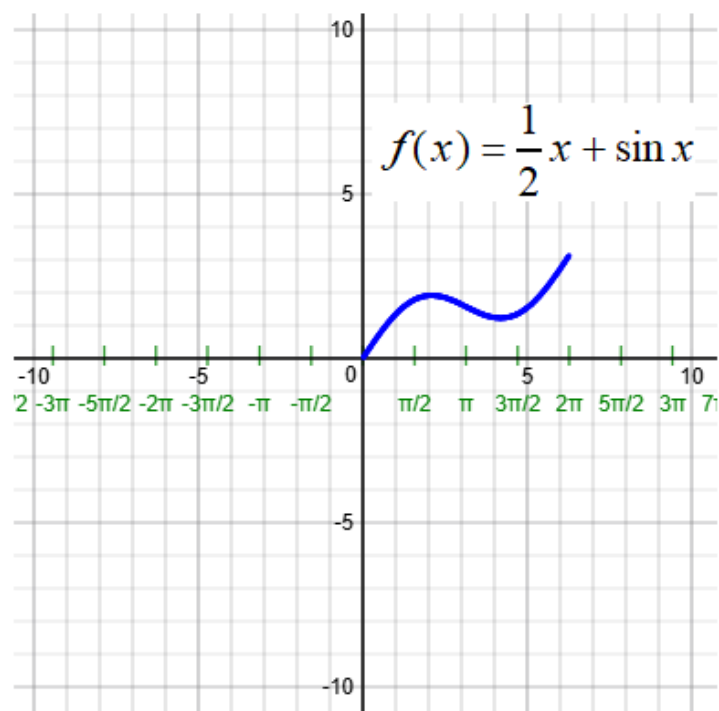
a) $f'(x) = \underline{\hspace{1cm}} ?$

b) $f''(x) = \underline{\hspace{1cm}} ?$

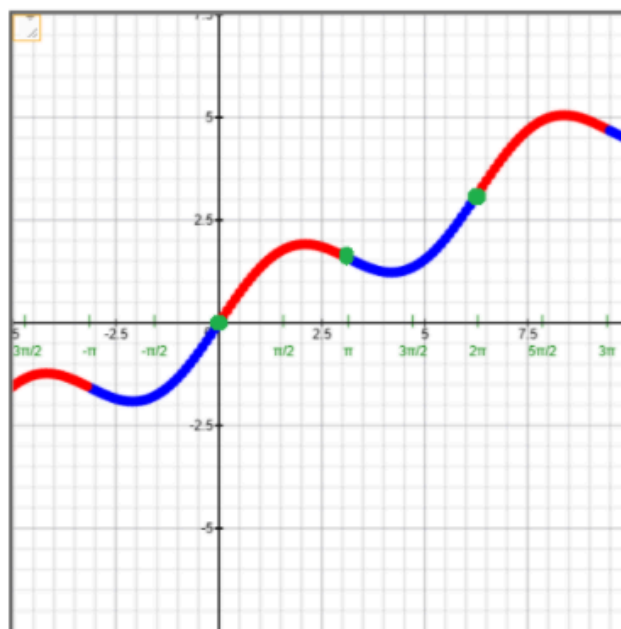
c) Set $f''(x) = 0$ and solve for x . $x = \underline{\hspace{1cm}} ?$

d) The function $f(x) = \frac{1}{2}x + \sin x$ concaves up on the interval: $\underline{\hspace{1cm}} ?$

e) The function $f(x) = \frac{1}{2}x + \sin x$ concaves down on the interval: $\underline{\hspace{1cm}} ?$



Concavity and Point of Inflection



● $f''(x) < 0$; 2nd derivative is negative

● $f''(x) > 0$; 2nd derivative is positive

● Point of Inflection

